

WEEK-1 graded assignment

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Q.2]

⇒ Since frequency $\propto \frac{1}{T}$

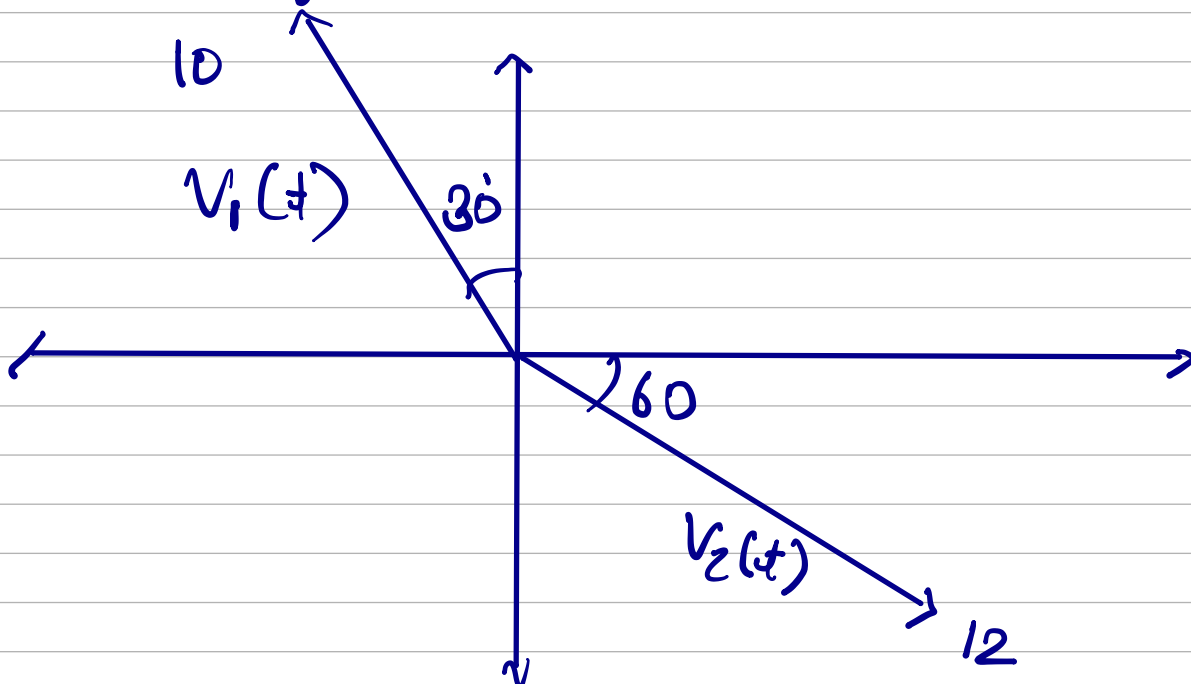
∴ more the time period less the frequency

more the frequency more faster will be change.

Q.3] $V_1(t) = 10 \cos(\omega t + 30^\circ)$

$V_2(t) = 12 \sin(\omega t - 60^\circ)$

Phasor diagram:-



Q.4]

Soln:- from phasor diagram $V_2(t)$ will lag & $V_2(t)$ is B

Q.5]

$$\Rightarrow V(t) = 10\sqrt{2} \sin(\omega t + 20^\circ)$$

$$\begin{aligned} \text{R.M.S value } V &= \frac{10\sqrt{2}}{\sqrt{2}} \\ &= 10 \end{aligned}$$

Q.6) Since reference signal is $\sqrt{2} \sin(\omega t)$

$\therefore 10\sqrt{2} \sin(\omega t + 20^\circ)$ is 20° leading

$$\therefore \angle 20^\circ$$

$$\text{Q.7) } V(t) = C + jD$$

$$V(t) = 10 \cos 20 + j 10 \sin 20$$

$$\therefore C = 10 \cos 20$$

$$= 9.39$$

$$\begin{aligned} Q.8) \quad \phi &= 10 \sin 20 \\ &= 3.420 \end{aligned}$$

Q.9)

$$\begin{aligned} \Rightarrow Z &= R + j\omega L \\ &= 10^3 + j 2\pi 10^4 \times 50 \times 10^{-3} \\ &= 10^3 + j \pi 10 \times 100 \\ &= 10^3 + j \pi 10^3 \end{aligned}$$

$$|Z| = \sqrt{10^6 + \pi^2 10^6}$$

$$|Z| = 10^3 \sqrt{1 + \pi^2}$$

$$\begin{aligned} |Z| &= 10^3 \times 3.29639 \\ &= 3295.3 \end{aligned}$$

Q.10)

$$\begin{aligned} \theta &= \tan^{-1} \left(\frac{XL}{R} \right) \\ &= \tan^{-1} \left(\frac{2\pi 10^4 \times 50 \times 10^{-3}}{10^3} \right) \end{aligned}$$

$$\theta = \tan^{-1} \left(\frac{\pi \cdot 10^3}{10^3} \right)$$

$$\theta = \tan^{-1}(3.14)$$

$$\theta = 72.33^\circ$$

Q.11)

$$I = \frac{V}{Z}$$

$$= \frac{10}{3295.3}$$

$$= 0.003034 \text{ A}$$

$$V_L = I \times X_L = 0.003034 \times 3140$$

$$V_L = 9.53 \text{ V}$$

Q.12) $V_R = I \times R$

$$= 0.003034 \times 1000$$

$$= 3.03 \text{ V}$$

Q.13)

⇒ we know for capacitive circuit
current leads voltage by 90°

$$\therefore \text{If } V(t) = V_m \sin(\omega t)$$

$$I(t) = I_m \sin(\omega t + 90^\circ)$$

Q.14)

$$\Rightarrow Z = \sqrt{R^2 + (X_L - X_C)^2}$$

we know at resonance $X_L = X_C$

$$\therefore X_L - X_C = 0$$

$$\therefore Z = R$$

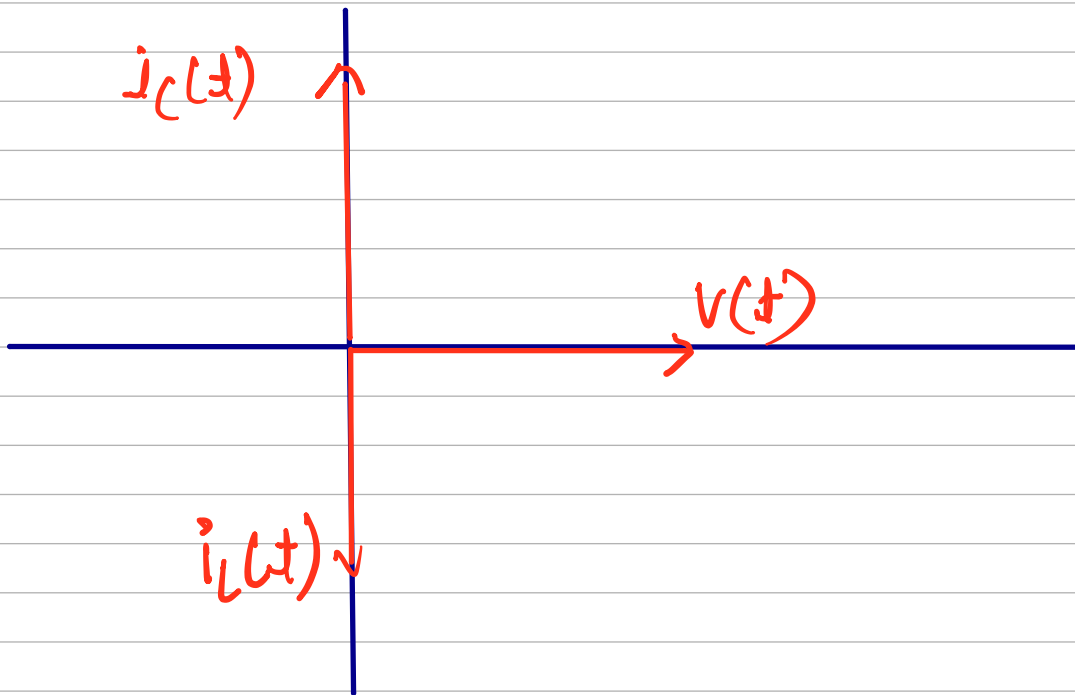
$\therefore$ Impedance is minimum as nothing
get added.

Q.15) $V(t) = 100 \sin(500t)$

$$L = 0.05 \text{ H}$$

$$i(t) = 2 \sin(500t + 90^\circ) \text{ A}$$

for LC circuit phasor representation will be



$$\text{Net current } i(t) = 2 \sin(500t + 90^\circ)$$

$$\text{Current} = \frac{V}{Z}$$

$$Z = \frac{V}{I}$$

$$= \frac{100}{2}$$

$$= 50 \Omega$$

$\therefore$ current is leading therefore
circuit is purely capacitive

$$\therefore Z = \sqrt{(X_L - X_C)^2}$$

$$Z = |X_L - X_C| = X_C - X_L$$

$$50 = \omega L - \frac{1}{\omega C}$$

$$\begin{aligned} \frac{1}{\omega C} &= 5000 \times 5 \times 10^{-2} - 50 \\ &= 50 \times 5 - 50 \end{aligned}$$

$$\frac{1}{\omega C} = 300$$

$$\begin{aligned} C &= \frac{1}{5000 \times 300} \\ &= \frac{1}{15 \times 10^5} \end{aligned}$$

$$= 0.666 \mu F$$

$$C = 0.67 \mu F$$

Q.16)

$$\Rightarrow V_R = 24 \text{ V}$$

$$V_L = 15 \text{ V}$$

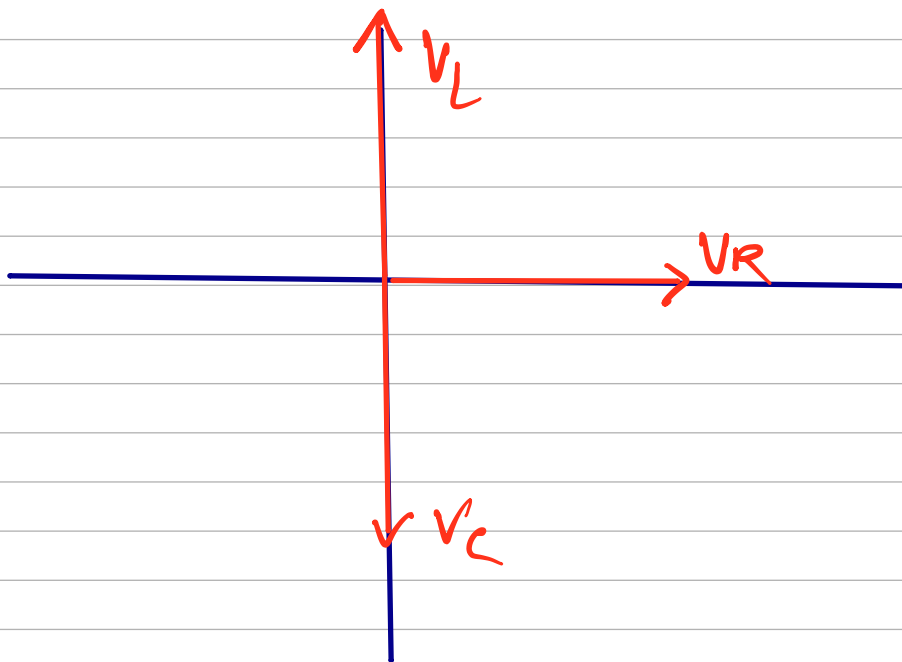
$$V_C = 45 \text{ V}$$

we know in RLC series circuit

V_R is phase with current

V_L will lead. by 90°

V_C will lag. by 90°



$$\therefore \text{Source voltage} = \sqrt{V_R^2 + (V_L - V_C)^2}$$

$$= \sqrt{24^2 + (30)^2}$$

$$= \sqrt{1476}$$

$$= 38.42 \text{ V}$$

Q.17)

Soln:-

$$V(t) = 100 \sin(2000t + 50^\circ) \text{ V}$$

$$i(t) = 20 \cos(2000t + 20^\circ) \text{ A.}$$

$\therefore$ Angle between voltage & current phasor $(110 - 50)$
 $= 60^\circ$

& current is leading by 60°

current is leading it means there is capacitive component as well as leading by less than 90° $\therefore$ RC circuit.

$$Q.18] \quad I = \frac{V}{Z}$$

$$|Z| = \frac{V}{I}$$

$$Z = \frac{100}{20}$$

$$Z = 5 \Omega$$

$$\text{phase angle} = 60^\circ$$

$$\therefore Z = R + jX_c$$

$$\therefore \text{Real component of } Z = 5 \cos 60^\circ$$

$$= \frac{5}{2}$$

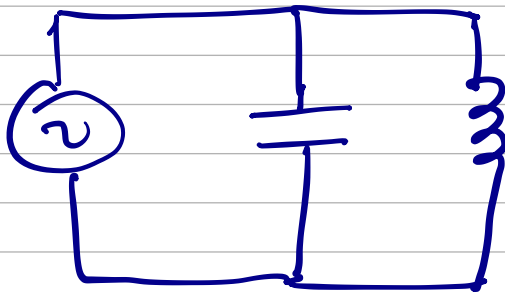
$$= 2.5 \Omega$$

Q.19]

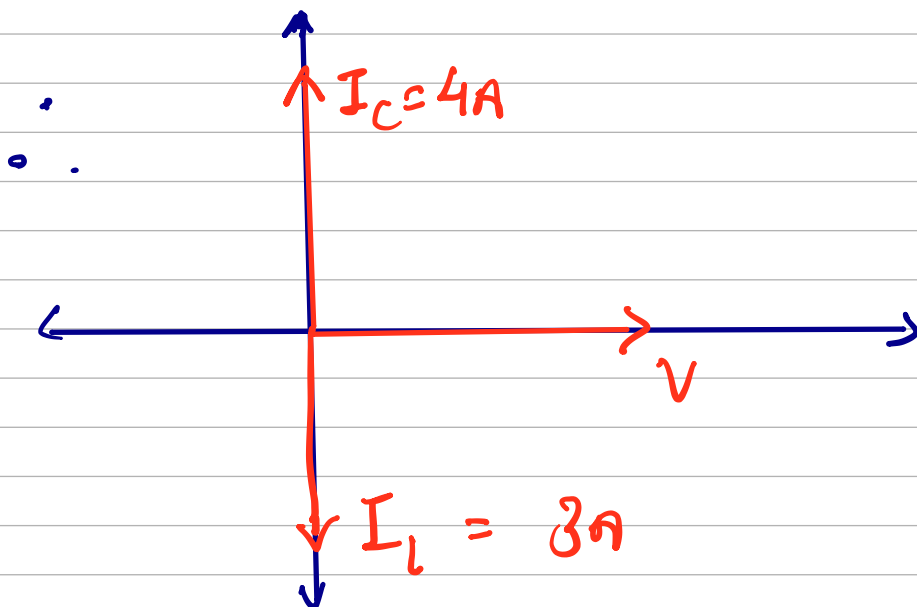
Soln:- frequency will remain unchanged for the circuit.

Q.20]

Soln:-



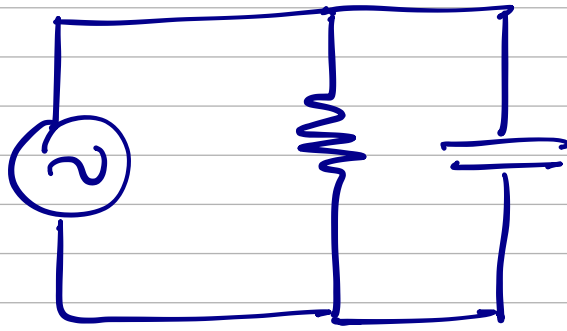
Voltage will be same.



$\therefore \text{Current} = I_C - I_L = 1A$

Q.21]

Solution



$$\therefore I_T = I_C + I_R$$

$$= 4 \angle 90^\circ + 3 \angle 0^\circ$$

$$I_T = \sqrt{4^2 + 3^2}$$

$$|I_T| = 5 \text{ A}$$